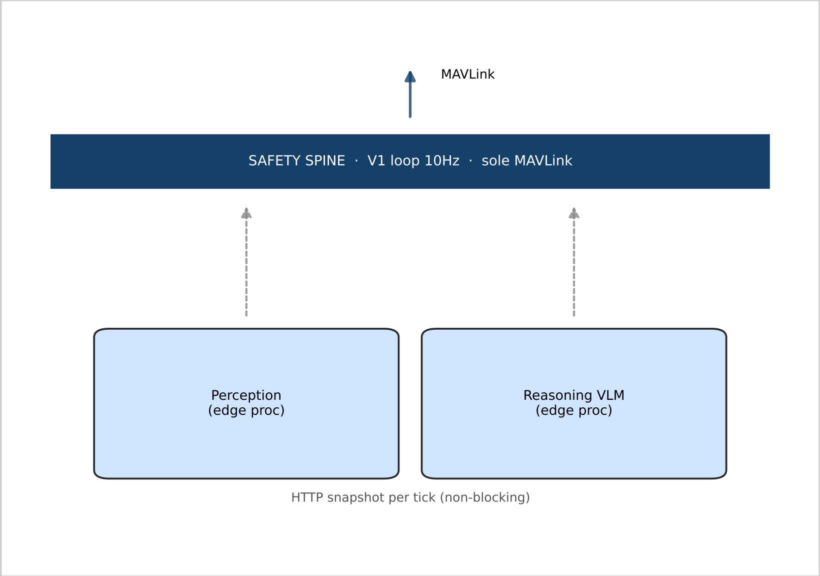
V2.0 Autonomy Architecture — Five Paradigms at a Glance

ConOps v0.3 + L0 v0.3.1 · judged on invariants

Extended-FSM

Sentinel-Spine

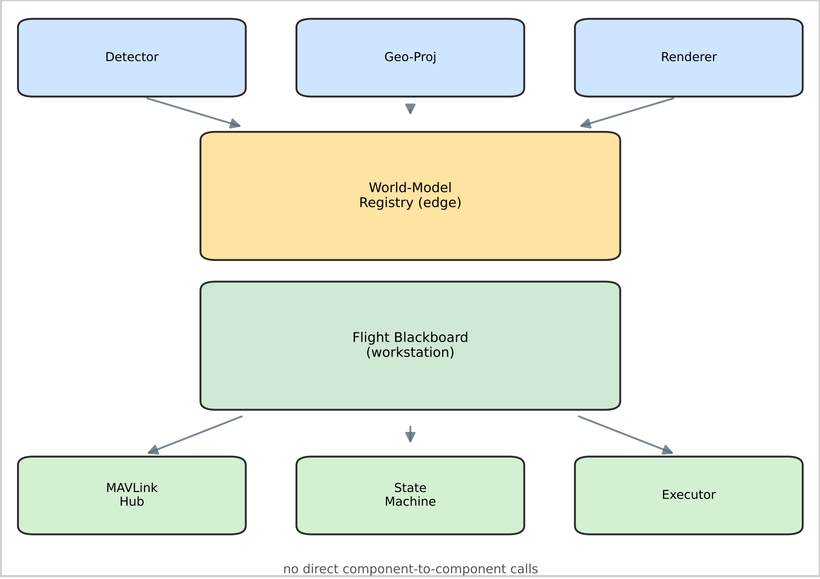
PASS



Blackboard

Skywriter

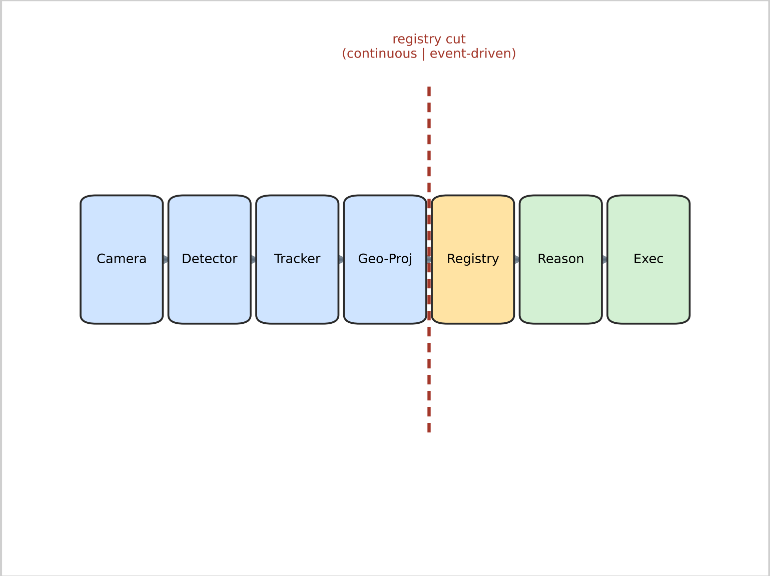
PASS



Dataflow

Streamline

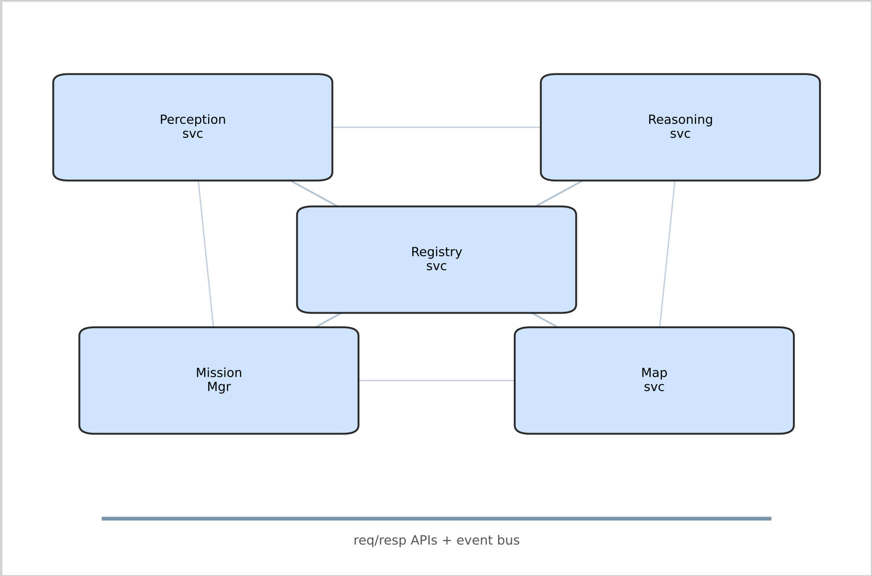
FAILS INVARIANT



Services

Sightline SOA

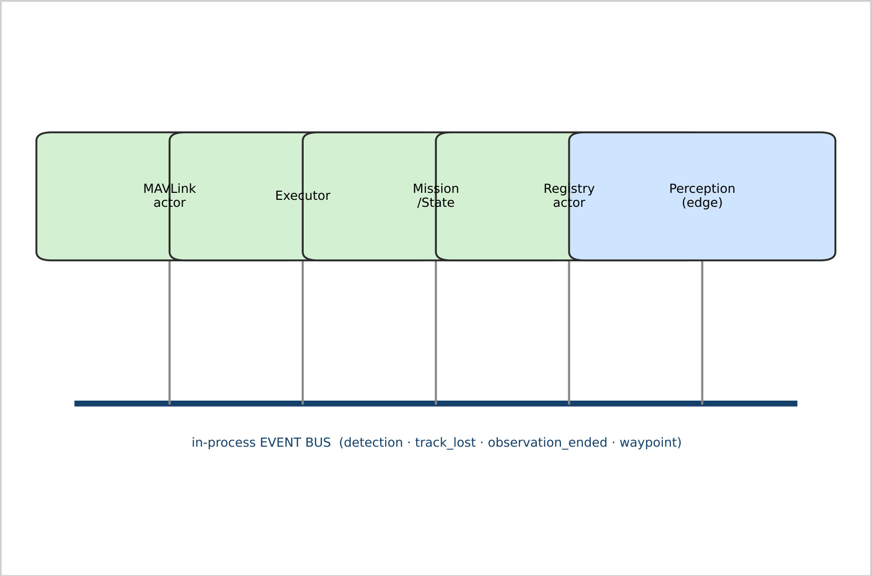
FAILS INVARIANT



Actor / bus

Aether Bus

FAILS INVARIANT



How They Differ — Structure & Coordination

	Sentinel-Spine Extended-FSM (PASS)	Skywriter Blackboard (PASS)	Streamline Dataflow (FAIL)	Sightline SOA Services (FAIL)	Aether Bus Actor / bus (FAIL)
Coordination mechanism	V1 single-owner 10 Hz spine; variable-latency work pushed to edge	Central blackboard(s); no direct component-to-component calls	Typed dataflow pipeline, cut at the registry	Independent services; req/resp APIs + event bus	Actors on an in-process event bus; 4 events drive wakeups
Where world-state lives	One in-process registry owned by the spine thread (workstation)	TWO blackboards in two fault domains: Registry (edge) + Flight (wkstn)	One registry as the pipeline cut-point (edge)	Registry on edge, in-process, exposed as a service	One Registry actor (workstation), message-driven
Perception <-> Reasoning	HTTP; spine ingests a perception snapshot each tick	Decoupled via blackboard reads / writes	Decoupled at the registry cut (continuous -> event)	Service APIs + event bus	Event bus + registry messages
Frame transport	Gazebo -> edge perception only; never reaches the spine	Edge shared-mem ring (drop-newest); stays on edge	In-process typed stream on edge (co-resident stages)	Direct sim -> edge net stream (RTSP/gRPC); bypasses MM	Terminates at the edge perception process
Gimbal command path	Spine executor over MAVLink (same channel as nav)	Intent as data; orbit loop on workstation via MAVLink	Owned by Executor (workstation) via MAVLink	Pointing driven direct to sim payload [IFC-01 breach]	Closed-loop pointing entirely on edge

How They Differ — Fault Behavior, Footprint & Verdict

	Sentinel-Spine Extended-FSM (PASS)	Skywriter Blackboard (PASS)	Streamline Dataflow (FAIL)	Sightline SOA Services (FAIL)	Aether Bus Actor / bus (FAIL)
If perception dies / domains	3 domains; perception + reasoning are edge procs; spine untouched	3 concentric firewalls; crash -> stale reads, mission continues	5 domains, registry = firewall; stall contained to writes	3 host-seam domains, but escalation rides the dying registry	3 nested; but safety actors + bus + both renderers in ONE proc
Footprint	14 components; smallest delta from proven V1	17 components / 2 hosts (heavy)	17 components	11 components, but many channels	14 components
Invariant verdict	PASS - zero violations	PASS - (overstates 'single owner' / '2 seams')	FAIL - IFC-01: 3rd I/O egress (gimbal channel)	FAIL - IFC-01 + SAF-02: perception commands gimbal/flight	FAIL - IFC-02: geo-proj crosses sim/real seam; SAF-07 abort unwired
Key strength / weakness	+ proven V1 safety loop, simplest / - spine = single point of failure; inherited on-thread sleeps	+ cannot-stall by construction / - replicated reads; edge-memory heavy	+ clean typed contracts / - registry contention; control logic bolted on	+ structural host firewall / - heaviest operational surface; self-defeating fault domain	+ clean actor isolation / - one wkstn proc = total-stop; sim/real story inconsistent

Recommendation — Sentinel-Spine (Extended-FSM)

the only zero-violation candidate

Adopt Sentinel-Spine, hardened with three grafts.

Only invariant-clean candidate; simplest; smallest delta from a verified-working V1 baseline. Its premise was checked against the V1 source (the VLM call is synchronous on the MAVLink-drain thread). Skywriter is also clean but over-engineered (17 components, two hosts, two blackboards) and over-pays the exact edge-memory contention the ConOps warns of. The other three disqualify themselves chasing distributed-systems elegance a single-Jetson research rig does not need.

GRAFT A - de-block the executor (most important pre-baseline fix)

The inherited V1 executor blocks the spine ~1s before every mode-set and ~16s on arm. Make mode-set non-blocking (issue, confirm on next telemetry tick; never sleep on the spine) - else per-tick orbit re-center stalls the drain.

GRAFT B - spine-resident observation watchdog (from a corrected Sightline)

A termination timer + feed-staleness watchdog on the spine that fires observation_ended / track_lost independent of the perception feed - so a whole-perception-domain death still terminates the orbit (closes SAF-08).

GRAFT C - first-class reasoning_ready event with overlap policy (from Aether Bus)

Coalesce/debounce per decision point: at most one in-flight reasoning request; a newer high-priority trigger supersedes a stale one; drop a reasoning_ready whose context is no longer current; safe default during the request window.

Honest residual: during an observation the orbit + gimbal close around a perception-derived position - sanctioned by TSK-02/03 and §6 (delegated control), not a SAF-02 breach - but the SE docs should say so plainly.

Open Decisions for You

before baselining the architecture

1. Edge-budget feasibility gate (SIM-01) - measure now.

Every survivor assumes perception + llama-server co-resident in 8GB. Run a resident-set measurement now. If it OOMs, the single-edge-box assumption breaks (smaller perception model / NVMe, or a 2nd edge unit - which erodes Sentinel-Spine's in-process-registry simplicity). This measurement could change the ranking.

2. Gimbal command path - confirm MAVLink.

Three candidates tripped IFC-01 by emitting gimbal pointing off-MAVLink. Confirm at L1 that gimbal/mount control rides MAVLink so IFC-02 'no change inside the stack' holds on real hardware.

3. Latency / threshold numbers (EVT-01, TSK-03/05, PER-06/07).

The architecture provides one gate seam for all of them; the numbers (TTL/staleness, persistence, confidence, slow-mover bound) must be set and validated against slow-mover dynamics before active pointing.

4. Spine concentration risk.

Registry + both renderers + observation controller now live in the one spine process - a bug there is a flight-safety bug, no hot standby (fine for SITL, not a real airframe). Decide internal module discipline and confirm RTL-fallback covers spine-internal faults.

5. Validate / reject / inconclusive criteria (ACT-01).

The actual research payload of V2. Decide who owns defining the three outcomes (prompt engineering + scoring harness); the held-out scorer (INT-01) depends on it.